



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Assignment-02 || Summer 2024

CSE 2233/CSI 233: Theory of Computation/Theory of Computing

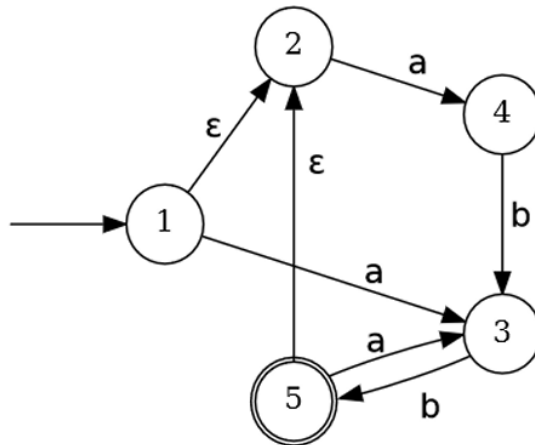
Total Marks: 10

Deadline: 28.09.24

Answer all questions. Figures are in the right-hand margin indicates full marks.

1. Consider the following- ϵ -NFA:

3

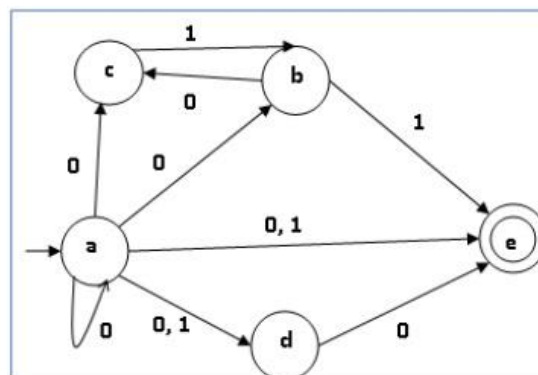


(a) Compute the ϵ -closure of each state.

(b) Convert the ϵ -NFA to equivalent DFA. Show both transition table and state diagram of the DFA.

2. Convert the following NFA over alphabet $\Sigma = \{0, 1\}$ to an equivalent DFA.

3



3. Convert the following regular expression into a finite automata:

4

- a) $L = \{w \mid w \text{ does not contain } xxx \text{ over the alphabet } \{x,y\}\}$
- b) $L = \{w \mid w \text{ starts with } b \text{ and contains at most } 3 \text{ a's over the alphabet } \{a, b, c\}\}$
- c) $L = \{w \mid w \text{ contains an even number of a's over the alphabet } \{a, b\}\}$
- d) $L = \{w \mid w \text{ contains not more than one occurrence of the substring "aa"}\}$